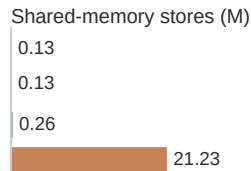
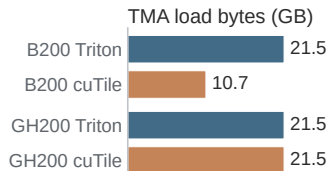
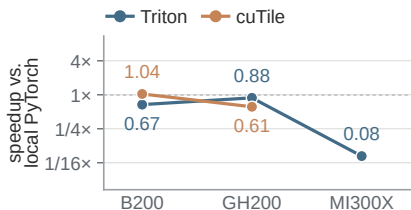


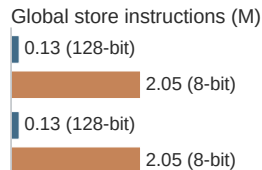
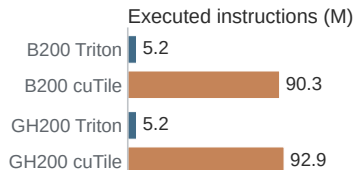
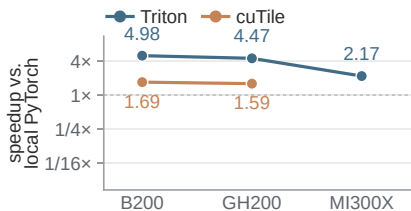
A Matrix operand delivery (matmul, FP32)



MI300X (static ISA): TensorDescriptor loads lowered to 96 scalar 32-bit global_load_dword; MFMA util. 4.5%

Confounders: Triton reads a B operand transposed before timing; cuTile tiles differ (256×256 on B200, 128×128 on GH200)

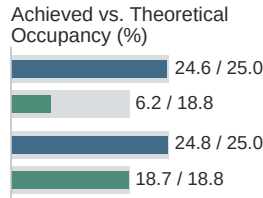
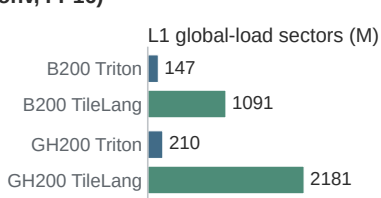
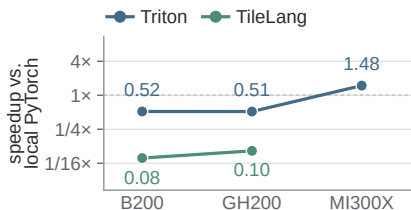
B Indexing overhead (destindex, INT8)



MI300X (static ISA): Triton stores each row with 4 per-lane buffer_store_byte, no vector store

Triton and cuTile write the same 2.05 M store sectors on both devices; the PyTorch baseline (index_copy) differs by vendor

C Memory access and latency hiding (1d_conv, FP16)



■ achieved ■ theoretical limit

MI300X: PyTorch runs MIOpen implicit GEMM plus 3 transpose kernels (trace); Triton loads with buffer_load_ushort (static ISA)

All loads are 16-bit and DRAM reads are equal (0.67 GB); TileLang uses a different kernel body on Hopper